

Q1: Write a MATLAB script to evaluate the function $f(x)$ for x values range between -7 to 7 with increment of 0.05 then calculate the value of x which produces the largest value for $f(x)$

$$f(x) = \begin{cases} (x-5)^2 & \text{if } x > 0 \\ -(x-3)^2 + 34 & \text{if } x \leq 0 \end{cases}$$

Q2: Solve the following differential equation and plot the result of the solution then label the axes.

$$(3 + 2y) \frac{dy}{dx} = 2 \cos(2x), y(0) = -1$$

Q3: Create the following matrices (MTX1 and MTX2)

$$MTX1 = \begin{bmatrix} 3 & 4 & 6 \\ -1 & 5 & 5 \\ 3 & -3 & 3 \end{bmatrix}, \quad MTX2 = \begin{bmatrix} 7 & 3 & 8 \\ 3 & 6 & 4 \\ -7 & 5 & 3 \end{bmatrix}$$

1. Write a code to find maximum and minimum numbers among MTX1 and MTX2.
2. Write a code to exchange column#2 of MTX1 with column#3 of MTX2.
3. Write a code to exchange row#1 of MTX1 with row#3 of MTX.